Two-Risky-Asset Markowitz Portfolio

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1 Introduction

This document summarizes the formulas and symbols used in the 2 risky-asset Markowitz portfolio theory. It includes expected return, standard deviation, and the relationship between asset weights.

2 Formulas

$$w_1 + w_2 = 1 \tag{1}$$

$$E(R_p) = w_1\mu_1 + w_2\mu_2 \tag{2}$$

$$\sigma_p = \sqrt{w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1w_2\rho\sigma_1\sigma_2} \tag{3}$$

3 Symbol Definitions

Symbol	Meaning
w_1	Weight of Asset 1 in the portfolio (fraction of total investment)
w_2	Weight of Asset 2 in the portfolio, $w_2 = 1 - w_1$
$E(R_1) = \mu_1$	Expected return of Asset 1
$E(R_2) = \mu_2$	Expected return of Asset 2
$E(R_p)$	Expected return of the portfolio
σ_1	Standard deviation (risk) of Asset 1
σ_2	Standard deviation (risk) of Asset 2
σ_p	Portfolio standard deviation (overall risk)
ρ	Correlation coefficient between Asset 1 and Asset 2 returns ($-1 \leq \rho \leq 1$)